

NexusForge — Detailed Product Requirements Document (PRD)

Version: 1.0 (May 2026)

Product Goal: A free, open-source, self-hostable AI workspace where any idea becomes a persistent, autonomous, multi-agent project that actually ships real work. It solves the core 2026 pain points: ideas get forgotten, agents hallucinate or lose context, and users waste time re-explaining everything. NexusForge gives users a living, self-improving workspace they can rely on daily.

Core Problem It Solves (Real User Pain in 2026)

- Agents forget context across sessions (“goldfish memory”).
- Single-prompt chatbots don’t complete complex, multi-step work.
- Users lose ideas in scattered chats, notes, or files.
- No easy way to turn vague seeds into shipped prototypes, code, or deliverables.
- Collaboration on AI-generated work is manual and error-prone.

NexusForge turns every seed into a **stateful, memory-rich, multi-agent organism** that works autonomously, remembers everything, uses real tools, and produces usable output.

1. Multimodal Seed Ingestion & Core Creation

What it does: Accepts any raw idea and instantly creates a persistent Nexus Core (the “organism”) with full initial context.

Input: Text prompt, voice recording, image(s), short video clip, or any combination (multimodal). Optional uploaded files (PDFs, docs, code repos).

Output: A new Nexus Core with:

- Unique ID and basic metadata.
- Initial NotebookLM-style DNA memory (embedded vector store of the seed + files).
- Spawned 5-agent swarm ready to work.

Real User Benefit: Users can start from messy real-world inputs (napkin sketch photo + voice note + PDF) and immediately get a structured, memory-rich project instead of copy-pasting into 5 different tools.

Why Useful: Eliminates the “how do I even start” friction that kills most ideas.

2. Persistent Multi-Tier Memory System (DNA Layer)

What it does: Maintains perfect long-term memory of the entire Nexus Core across sessions, devices, and time.

Input: Any interaction (chat, file upload, agent output, user edit).

Output:

- All data automatically embedded and stored in a personal vector database.
- Agents always retrieve relevant past context before reasoning.
- Full searchable history of every version, decision, and file.

Real User Benefit: The organism “knows” everything it has ever done or been told. No more re-explaining requirements.

Why Useful: Solves the #1 2026 agent complaint (memory loss). Users treat it like a true second brain.

3. Stateful Multi-Agent Swarm Execution

What it does: Runs a dedicated crew of 5 specialized agents that collaborate as a team to develop the Nexus Core.

Input: User goal or message (e.g., “Turn this into a working web app”).

Output: Agents autonomously:

- Plan subtasks.
- Debate options (visible in chat).
- Generate code, documents, visuals, or prototypes.
- Critic agent verifies output for accuracy before finalizing.

Real User Benefit: The swarm completes complex, multi-step work instead of giving one-off answers.

Why Useful: Mirrors proven 2026 patterns (LangGraph/CrewAI) where agents actually finish tasks instead of just chatting.

4. Tool Use & Secure Sandbox Execution

What it does: Agents can take real actions using safe, sandboxed tools instead of just

describing them.

Input: Any task that requires external action (code running, web research, file creation).

Output:

- Builder agent runs code in isolated sandbox and returns results.
- Agents call APIs (GitHub, etc.) with user approval gates.
- Generated artifacts (working code, exported files) are stored inside the Nexus Core.

Real User Benefit: Users get runnable prototypes, tested code, or processed files — not just text suggestions.

Why Useful: Turns ideas into shippable deliverables (the biggest gap in current chat-based tools).

5. Autonomous Background Evolution & Mutation

What it does: The Nexus Core continues improving itself in the background without user input.

Input: None (scheduled).

Output:

- Periodic mutations (new versions, research summaries, code improvements).
- Changelog and impact summary when user returns.
- Optional user-defined triggers (e.g., “evolve every 6 hours”).

Real User Benefit: Progress happens while you sleep or work on other things. You return to an evolved project.

Why Useful: Real 2026 agentic value — agents that keep working when you’re offline.

6. Semantic Forking & Safe Collaboration

What it does: Enables controlled sharing and merging of Nexus Cores.

Input: User marks a Core as shareable (private link or public opt-in).

Output:

- Semantic search finds similar public Cores.
- One-click fork creates an independent copy with merged DNA where chosen.

- Real-time multi-user editing (changes sync instantly).

Real User Benefit: Users can safely build on others' work or collaborate without losing control of their original Core.

Why Useful: Creates network effects while protecting intellectual property (critical for real adoption).

7. Version History with Intelligent Diffs & Rollback

What it does: Tracks every change to the Nexus Core with AI understanding.

Input: Any mutation or user edit.

Output:

- Complete version history.
- Gemini-generated natural-language diffs ("This version added authentication and improved UX by X").
- One-click rollback to any previous state.

Real User Benefit: Users can experiment freely and always revert safely.

Why Useful: Prevents irreversible mistakes and gives clear audit trail for teams.

8. One-Click Export & Delivery

What it does: Converts any version of the Nexus Core into real-world deliverables.

Input: User selects export type.

Output:

- GitHub-ready repo (code + README + tests).
- PDF or Markdown summary with all context.
- Structured data exports (JSON, CSV).
- Deploy-ready artifacts (if applicable).

Real User Benefit: Users get something they can immediately use or ship instead of manual copying.

Why Useful: Closes the loop from idea → finished output.

9. Personal RAG Knowledge Base

What it does: Users upload their own files/repos/notes; the swarm automatically grounds all reasoning in them.

Input: Uploaded personal documents.

Output: Agents reference user data in every response and evolution cycle.

Real User Benefit: The organism becomes truly personalized and accurate to the user's world.

Why Useful: Eliminates generic hallucinations when working on private projects.

10. Self-Reflection & Impact Reporting

What it does: The swarm regularly evaluates its own progress.

Input: None (automatic after major cycles).

Output: Weekly or on-demand report: "This Core evolved in these ways, delivered X value, with these remaining risks."

Real User Benefit: Users get objective feedback on progress and can course-correct early.

Why Useful: Builds trust and prevents wasted effort on dead-end paths.

These 10 features are **strictly practical** and directly address 2026 user needs identified across tools like Cursor, Claude Projects, Mem, LangGraph deployments, and enterprise agent platforms. They produce measurable output (code, documents, prototypes, decisions) that users can actually use and ship — not just pretty visuals.

The entire system is designed to be **free & open-source**, self-hostable, and run on the user's own Gemini key so there is no vendor lock-in or surprise costs.

This PRD is ready to hand to your development team or Claude. It focuses exclusively on functionality that delivers real daily value.

If you want any feature expanded with acceptance criteria or technical notes, let me know.